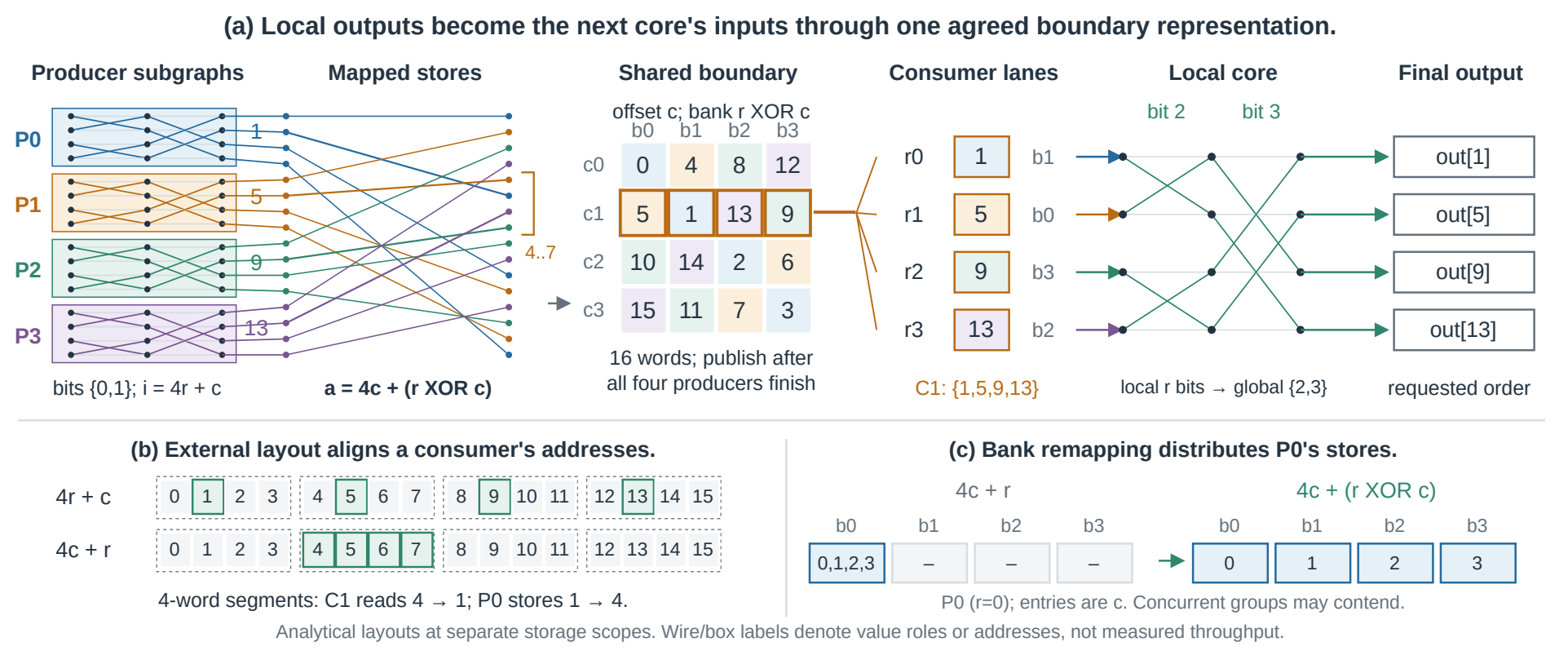
(a) Local outputs become the next core's inputs through one agreed boundary representation.
Producer subgraphs
Mapped stores
Shared boundary
Consumer lanes
Local core
Final output
P0
P1
P2
P3
1
5
9
13
4..7
bits {0,1}; i = 4r + c
a = 4c + (r XOR c)
offset c; bank r XOR c
b0 b1 b2 b3
c0 0 4 8 12
c1 5 1 13 9
c2 10 14 2 6
c3 15 11 7 3
16 words; publish after all four producers finish
r0 1 b1
r1 5 b0
r2 9 b3
r3 13 b2
C1: {1,5,9,13}
bit 2
bit 3
out[1]
out[5]
out[9]
out[13]
local r bits → global {2,3}
requested order
(b) External layout aligns a consumer's addresses.
4r + c
4c + r
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
4-word segments: C1 reads 4 → 1; P0 stores 1 → 4.
(c) Bank remapping distributes P0's stores.
4c + r
4c + (r XOR c)
b0 b1 b2 b3
0,1,2,3 – – –
0 1 2 3
P0 (r=0); entries are c. Concurrent groups may contend.
Analytical layouts at separate storage scopes. Wire/box labels denote value roles or addresses, not measured throughput.